

Applied Statistics (MATH10096)

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The exponential family of distributions

- ↪ In a GLM, the response variable can follow any distribution from the **exponential family**.
- ↪ A distribution belongs to the exponential family of distributions if its probability density function, or probability mass function, can be written as

$$f(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\}.$$

- ↪ Here:
 - ↪ θ is known as the **canonical parameter** and is a function of the expectation of the response variable, $\mu \equiv \mathbb{E}(Y)$, and so θ is completely determined by β .
 - ↪ $\phi > 0$ is called the **dispersion parameter**. It may or may not be known and represents the scale.
 - ↪ $a(\cdot)$, $b(\cdot)$, and $c(\cdot)$ are known functions that differ across members of the exponential family.

The exponential family of distributions

Poisson distribution

↪ The probability mass function of a Poisson distribution is

$$f(y; \mu) = \frac{e^{-\mu} \mu^y}{y!}, \quad y = 0, 1, 2, \dots, \quad \mu = \mathbb{E}(Y) > 0.$$

↪ Let us first write $f(y; \mu)$ as $\exp\{\log f(y; \mu)\}$. We have that

$$\log f(y; \mu) = -\mu + y \log(\mu) - \log(y!) \Rightarrow \exp\{\log f(y; \mu)\} = \exp\{-\mu + y \log(\mu) - \log(y!)\}.$$

↪ Now, let us rearrange the equation in the exponential family of distributions form

$$\exp\{\log f(y; \mu)\} = \left\{ \frac{y \log(\mu) - \mu}{1} + [-\log(y!)] \right\}.$$

The exponential family of distributions

Poisson distribution

↪ We have

$$\exp\{\log f(y; \mu)\} = \left\{ \frac{y \log(\mu) - \mu}{1} + [-\log(y!)] \right\}.$$

↪ This is in the exponential family of distributions with:

$$\hookrightarrow \theta = \log \mu \Rightarrow \mu = e^\theta$$

$$\hookrightarrow b(\theta) = \mu = e^\theta$$

$$\hookrightarrow a(\phi) = \phi = 1$$

$$\hookrightarrow c(y, \phi) = -\log(y!)$$

The exponential family of distributions

Gaussian distribution

↪ The probability density function of a Gaussian distribution with mean μ and variance σ^2 is given by

$$f(y; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (y - \mu)^2 \right\}, \quad y \in \mathbb{R}, \quad \mu \in \mathbb{R}, \quad \sigma > 0.$$

↪ As before, writing $f(y; \mu, \sigma^2)$ as $\exp\{\log f(y; \mu, \sigma^2)\}$:

$$\begin{aligned} \log f(y; \mu, \sigma^2) &= -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (y - \mu)^2 \\ &= -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (y^2 - 2\mu y + \mu^2), \end{aligned}$$

and thus

$$\begin{aligned} \exp\{\log f(y; \mu, \sigma^2)\} &= \exp \left\{ -\frac{1}{2\sigma^2} (y^2 - 2\mu y + \mu^2) - \frac{1}{2} \log(2\pi\sigma^2) \right\} \\ &= \exp \left\{ \frac{y\mu - \mu^2/2}{\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) - \frac{y^2}{2\sigma^2} \right\} \end{aligned}$$

The exponential family of distributions

Gaussian distribution

↪ We have

$$\exp \left\{ \frac{y\mu - \mu^2/2}{\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) - \frac{y^2}{2\sigma^2} \right\}.$$

↪ This is in the exponential family of distributions with:

$$\hookrightarrow \theta = \mu$$

$$\hookrightarrow b(\theta) = \frac{\mu^2}{2} = \frac{\theta^2}{2}$$

$$\hookrightarrow a(\phi) = \phi = \sigma^2$$

$$\hookrightarrow c(y, \phi) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{y^2}{2\sigma^2} = -\frac{1}{2} \log(2\pi\phi) - \frac{y^2}{2\phi}$$

The exponential family of distributions

Gamma distribution

→ The density of the Gamma distribution is usually given by

$$f(y; \nu, \lambda) = \frac{\lambda^\nu}{\Gamma(\nu)} y^{\nu-1} e^{-\lambda y}, \quad y > 0, \quad \lambda > 0, \quad \nu > 0,$$

→ Here ν describes the shape and λ the rate (inverse of a scale parameter) of the distribution.

→ Under this parametrization, $E(Y) \equiv \mu = \frac{\nu}{\lambda}$.

→ However, for the purposes of GLM, it is convenient to reparametrize the distribution by its mean, which can be achieved by letting $\lambda = \frac{\nu}{\mu}$, leading to

$$f(y; \nu, \mu) = \frac{1}{\Gamma(\nu)} \left(\frac{\nu}{\mu} \right)^\nu y^{\nu-1} e^{-\frac{\nu}{\mu} y}.$$

→ Note that now $E(Y) = \mu$.

The exponential family of distributions

Gamma distribution

↪ As for the other two examples,

$$\log f(y; \nu, \mu) = -\log[\Gamma(\nu)] + \nu \log(\nu) - \nu \log(\mu) + (\nu - 1) \log(y) - \frac{\nu y}{\mu}.$$

Hence,

$$\begin{aligned} \exp\{\log f(y; \nu, \mu)\} &= \exp\left\{-\frac{\nu y}{\mu} - \nu \log(\mu) + (-\log[\Gamma(\nu)] + \nu \log(\nu) + (\nu - 1) \log(y))\right\} \\ &= \exp\left\{\frac{y(-1/\mu) - \log(\mu)}{1/\nu} + (-\log[\Gamma(\nu)] + \nu \log(\nu) + (\nu - 1) \log(y))\right\}. \end{aligned}$$

↪ This is in the exponential family of distributions with:

$$\hookrightarrow \theta = -\frac{1}{\mu} \Rightarrow \mu = -\frac{1}{\theta}$$

$$\hookrightarrow b(\theta) = \log(\mu) = \log(-\frac{1}{\theta}) = -\log(-\theta)$$

$$\hookrightarrow a(\phi) = \phi = \frac{1}{\nu}$$

$$\hookrightarrow c(y, \phi) = -\log[\Gamma(\nu)] + \nu \log(\nu) + (\nu - 1) \log(y) = -\log[\Gamma(1/\phi)y] + \frac{1}{\phi} \log\left(\frac{y}{\phi}\right)$$

Mean of the exponential family of distributions

- ↪ We now derive the general expressions for the **mean and variance of the exponential family of distributions**.
- ↪ To this end, note that the **log likelihood of θ** , for a single observation y , is simply $\log f(y; \theta, \phi)$ considered as a function of θ , i.e.,

$$l(\theta, \phi) = \log f(y; \theta, \phi) = \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi).$$

- ↪ Taking derivatives with respect to θ gives

$$\frac{\partial l}{\partial \theta} = \frac{y - b'(\theta)}{a(\phi)},$$

- ↪ Treating l as a random variable, by replacing the particular observation y by the random variable Y , enables the expected value of $\partial l / \partial \theta$ to be evaluated

$$\mathbb{E} \left(\frac{\partial l}{\partial \theta} \right) = \mathbb{E} \left(\frac{Y - b'(\theta)}{a(\phi)} \right) = \frac{\mathbb{E}(Y) - b'(\theta)}{a(\phi)}.$$

Mean of the exponential family of distributions

↪ From general likelihood theory, it is known that

$$\mathbb{E} \left(\frac{\partial l}{\partial \theta} \right) = 0.$$

↪ This result implies that

$$\frac{\mathbb{E}(Y) - b'(\theta)}{a(\phi)} = 0 \Rightarrow \mu \equiv E(Y) = b'(\theta). \quad (1)$$

- ↪ Note that $b'(\theta)$ is one-to-one, so there is a one to one correspondence between μ and θ .
- ↪ The result in (1) is the key to linking the GLM model parameter β to the canonical parameter of the exponential family θ .
- ↪ In a GLM, β determines the mean of the response variable and via, Equation (1), thereby determines the canonical parameter for each response observation.

Variance of the exponential family of distributions

↪ We now proceed to determine $\text{var}(Y)$.

↪ Taking second derivatives of the log likelihood, we obtain

$$\frac{\partial^2 l}{\partial \theta^2} = -\frac{b''(\theta)}{a(\phi)}.$$

↪ General likelihood theory also provides the following result

$$\mathbb{E} \left(\frac{\partial^2 l}{\partial \theta^2} \right) = -\mathbb{E} \left[\left(\frac{\partial l}{\partial \theta} \right)^2 \right].$$

Variance of the exponential family of distributions

↪ First, note that

$$\mathbb{E} \left(\frac{\partial^2 l}{\partial \theta^2} \right) = \mathbb{E} \left(-\frac{b''(\theta)}{a(\phi)} \right) = -\frac{b''(\theta)}{a(\phi)}.$$

↪ Additionally, note that

$$\left(\frac{\partial l}{\partial \theta} \right)^2 = \left(\frac{y - b'(\theta)}{a(\phi)} \right)^2 = \frac{(y - b'(\theta))^2}{a^2(\phi)} = \frac{(y - \mu)^2}{a^2(\phi)},$$

leading to

$$\mathbb{E} \left[\left(\frac{\partial l}{\partial \theta} \right)^2 \right] = \mathbb{E} \left(\frac{(Y - \mu)^2}{a^2(\phi)} \right) = \frac{\mathbb{E}((Y - \mu)^2)}{a^2(\phi)} = \frac{\text{var}(Y)}{a^2(\phi)}.$$

Variance of the exponential family of distributions

↪ The general result,

$$\mathbb{E}\left(\frac{\partial^2 l}{\partial \theta^2}\right) = -\mathbb{E}\left[\left(\frac{\partial l}{\partial \theta}\right)^2\right],$$

implies that

$$-\frac{b''(\theta)}{a(\phi)} = -\frac{\text{var}(Y)}{a^2(\phi)},$$

↪ Therefore

$$\text{var}(Y) = b''(\theta)a(\phi).$$

↪ We see that the mean is a function of θ only, while the variance is a product of functions of the location and scale.

Exponential family of distributions

Variance function

- ↪ In principle, a could be any function of ϕ .
- ↪ However, the choice of $a(\phi) = \phi/\omega$, where ϕ is constant over all observations and ω is a known weight that may vary from observation to observation, encompasses all practical cases needed.
- ↪ Hence, we now have

$$\text{var}(Y) = b''(\theta) \frac{\phi}{\omega} = V(\mu)\phi.$$

- ↪ The function $V(\mu) = b''(\theta)/\omega$ is called the **variance function** and describes how the variance relates to the mean using the known relationship between θ and μ (through Equation (1)).

Exponential family of distributions

Variance function: Gaussian distribution

↪ In the Gaussian case, we had

$$\omega = 1 \quad \text{and} \quad b(\theta) = \theta^2/2,$$

implying that

$$V(\mu) = b''(\theta) = 1,$$

and so the variance is independent of the mean.

↪ For other distributions this is not true, making the Gaussian case exceptional.

↪ Further, observe that $\text{var}(Y) = 1 \times \phi = \sigma^2$, as we already knew.

Exponential family of distributions

Variance function: Poisson distribution

↪ For the Poisson distribution, we had

$$\omega = 1 \quad \text{and} \quad b(\theta) = e^{\theta},$$

implying that

$$b''(\theta) = e^{\theta} = \mu,$$

↪ Thus $V(\mu) = \mu$, i.e., the variance function is linear.

↪ Also, $\text{var}(Y) = \phi \times \mu = \mu$ as it should be (it is widely known that for the Poisson distribution the mean coincides with the variance).

Exponential family of distributions

Variance function: Gamma distribution

↪ For the Gamma distribution we had that

$$\omega = 1, \quad \text{and} \quad b(\theta) = -\log(-\theta) \Rightarrow b''(\theta) = 1/\theta^2,$$

and thus $V(\mu) = \mu^2$, that is, the variance function is a quadratic function of the mean.

↪ Lastly, $\text{var}(Y) = b''(\theta)\phi = \mu^2/\nu$.

Exponential family of distributions

Link function

- ↪ The **link function** of a GLM connects the random component to the linear predictor.
- ↪ That is, a GLM states that the linear predictor for observation i , $\eta_i = \mathbf{x}_i^T \beta = \sum_{j=1}^p \beta_j x_{ij}$ relates to μ_i by $\eta_i = g(\mu_i)$, for a link function g .
- ↪ Equivalently, $\mu_i = g^{-1}(\eta_i)$, i.e., the function g^{-1} maps linear predictor values to the mean.

Exponential family of distributions

Link function

- ↪ By allowing nonlinear relationships between the mean response and the predictors, the link function enables the relationship between the linear predictor and the mean response to be chosen based on subject-matter or statistical considerations.
- ↪ Restrictions on μ_i can be imposed through the choice of g , which maps the domain of μ_i to the set inhabited by η_i , usually the real line.
- ↪ In principle, any smooth monotonic function g will do, but there are some convenient and common choices for the standard GLMs.

Exponential family of distributions

Link function

↪ The usual link function for the normal distribution is the identity link, $\eta_i = g(\mu_i) = \mu_i$, yielding the normal linear model we covered earlier in the course.

↪ Suppose instead that $y_i = \mu_i + \epsilon_i$, but

$$\mu_i = \frac{\alpha_0 x_i}{\alpha_1 + x_i}.$$

↪ On rewriting,

$$\mu_i^{-1} = \frac{\alpha_1 + x_i}{\alpha_0 x_i} = \frac{1}{\alpha_0} + \frac{\alpha_1}{\alpha_0} \frac{1}{x_i} = \beta_0 + \beta_1 z_i, \quad z_i = \frac{1}{x_i}, \quad \beta_0 = \frac{1}{\alpha_0}, \quad \beta_1 = \frac{\alpha_1}{\alpha_0}.$$

↪ We see that this model falls into the general GLM setup with normal distribution for the response and the inverse link function, $\eta_i = g(\mu_i) = \mu_i^{-1}$.

Exponential family of distributions

Link function

- ↪ For the Poisson GLM, the mean μ_i must be positive, so $\eta_i = \mu_i$ will not work conveniently since η_i can be negative.
- ↪ The standard choice is $\mu_i = e^{\eta_i}$, so that $\eta_i = \log \mu_i$, which ensures $\mu_i > 0$.
- ↪ This log link means that additive effects of the predictors $\mathbf{x}_i$ lead to multiplicative effects on the mean response μ_i .

Exponential family of distributions

Link function

- ↪ The link function g , which transforms the mean μ_i to the canonical parameter θ_i in the definition of the exponential family, is called the **canonical link function**.
- ↪ That is, g is such that

$$\eta_i = g(\mu_i) = \theta_i.$$

- ↪ Since $b'(\theta_i) = \mu_i$ and b' is a one to one function, it follows that $\theta_i = b'^{-1}(\mu_i)$.
- ↪ Thus, the canonical link function is given by

$$g(\mu_i) = b'^{-1}(\mu_i).$$

Exponential family of distributions

Link function

- ↪ The canonical link for the Gaussian distribution is $g(\mu) = \mu$.
- ↪ For the Poisson distribution it is $g(\mu) = \log \mu$.
- ↪ For the Gamma distribution, $\theta = -1/\mu$ and the canonical link is $g(\mu) = -1/\mu$.
- ↪ Note that for the gamma distribution, $\mu > 0$, and therefore, $g(\mu) = -1/\mu$ takes values in $(-\infty, 0)$, which does not match the range of the linear predictor.
- ↪ The log link function may be more appropriate, as it at least matches the range of the predictor.